EDSL EXPERIMENT REPORT

Talk Isn't Always Cheap

A durable multi-agent debate pilot

Three GPT-4o-mini agents × one CommonSenseQA item × three stages

Local EDSL orchestration; direct OpenAI inference; SQLite workflow and shared-state audit trail

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Executive summary

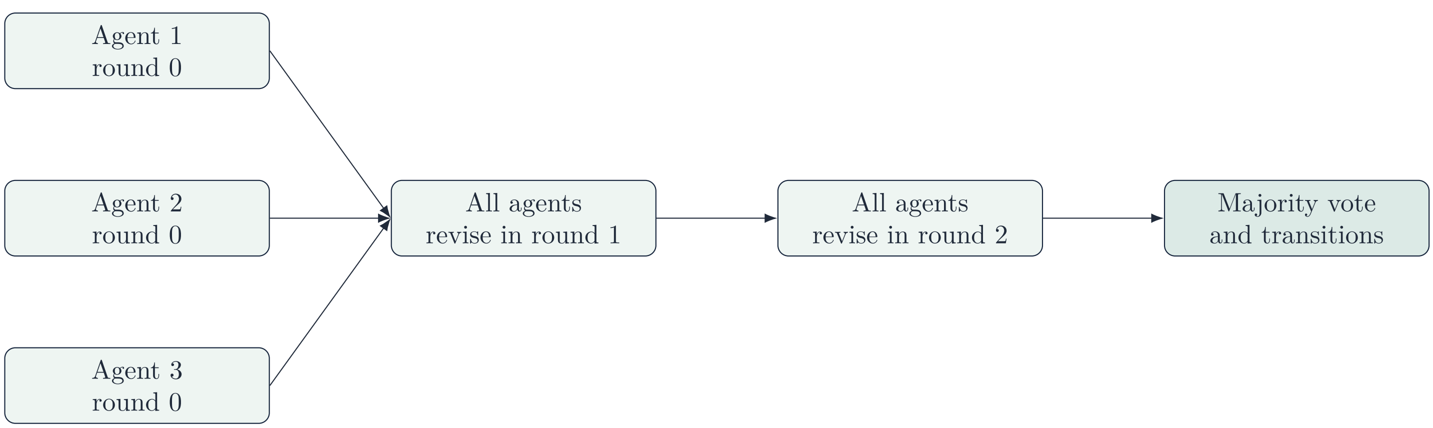
This report documents a small, runnable EDSL design replication of Wynn, Satija, and Hadfield's *Talk Isn't Always Cheap: Understanding Failure Modes in Multi-Agent Debate* (arXiv:2509.05396v2). The original paper studies when exchanging reasons causes groups of language models to become less accurate. Our implementation expresses the protocol as a serializable EDSL workflow, records every response in transactional shared state, and computes majority and participant-level answer transitions deterministically.

> **Pilot result.** All three GPT-4o-mini agents independently selected option B, "wearing out," although the experiment fixture records D, "falling apart," as the reference answer. Every agent retained B in both debate rounds. The group was therefore incorrect before and after debate, with six *incorrect* → *incorrect* transitions and no answer flips.

This is a successful execution test, not a statistical replication. It covers one item, one seed, one homogeneous model composition, and the base prompt. The paper's principal study covers 100 items from each of three datasets, five seeds, and ten homogeneous or heterogeneous three-agent compositions. The pilot cannot establish whether debate generally helps or harms accuracy.

1 Original experiment

Each group contains three agents. At round 0 each answers independently and supplies reasoning. At debate rounds 1 and 2, each agent receives its previous response and the other agents' preceding responses, then revises its answer. The final group answer is the majority vote.



The paper evaluates CommonSenseQA (CSQA), MMLU, and GSM8K using GPT-4o-mini, LLaMA-3.1-8B-Instruct, and Mistral-7B-Instruct-v0.2. It uses default temperature, `top_p=0.9`, a 2,048-token generation ceiling, 100 sampled items per dataset, and five seeds. It also compares the base prompt with a correctness-payoff intervention.

For the directly comparable homogeneous GPT condition on CSQA, the paper reports 75.6% accuracy without debate and 74.8% after debate, a 0.8 percentage-point decline. Across configurations, the paper's central evidence includes declining accuracy over rounds, more correct-to-incorrect than incorrect-to-correct changes in important settings, and sensitivity to how many peers agree.

2 EDSL architecture

The implementation separates three concerns:

1. **Workflow**: creates three synchronized fan-out stages. A stage completes only after all three participants submit, so no participant sees same-round answers.

2. **Shared state**: appends each participant, round, answer, and rationale to a durable ledger. Writes are versioned and idempotent in SQLite.
3. **Analysis**: derives majority correctness and the four transition classes from saved records, rather than asking a language model to score itself.

2.1 A typed experiment item

```
EXAMPLE_ITEM = DebateItem(
    item_id="csqa-example",
    dataset="csqa",
    question="If a product doesn't last, what does it have a reputation of doing?",
    options=("A) disintegrating", "B) wearing out", "C) dissolving",
             "D) falling apart", "E) dissipating"),
    correct_answer="D",
)
```

CSQA and MMLU items require labeled options and a valid answer code. GSM8K remains open-ended: its item has no options and requests only a final numerical answer. This distinction prevents the workflow from silently converting the math benchmark into multiple choice.

2.2 One model call returns answer and reasoning

```
response = QuestionDict(
    question_name="response",
    question_text=answer_text + answer_format
        + " Provide concise reasoning and do not merely appeal to consensus.",
    answer_keys=["answer", "reasoning"],
    value_types=["str", "str"],
    value_descriptions=[
        "The selected answer in the required format.",
        "A concise bullet-point summary of the supporting reasoning.",
    ],
    include_comment=False,
)
```

Using one structured question per participant-stage matches the paper's one-generation response more closely than separate answer and rationale questions. It also yields exactly nine calls in this pilot: three participants times three stages.

2.3 Append-only shared state

```
DEBATE_LEDGER = Machine(
    name="DebateLedger",
    constants={},
    fields={"responses": state_field(T.sequence(), [])},
    commands={
        "submit": Command(
            inputs={"participant": T.text(),
                    "round": T.integer(minimum=0, maximum=2),
                    "response": T.map()},
            effects=(append("responses", record(
                participant=input_("participant"),
                round=input_("round"),
                response=input_("response"))),),
        )
    },
    view={"responses": field("responses")},
)
```

The workflow connects each answer to the ledger using symbolic references. The runtime resolves `current.agent.name` and `response.answer` only after a validated submission exists.

```
step = builder.step(
    f"round-{round_number}", Survey([response]),
    assigned_to=role("debater"), after=previous,
    visible_to=role("debater"),
    reads=(() if previous is None else (ledger.read(),)),
    writes=(ledger.submit(
        participant=current.agent.name,
        round=round_number,
```

```
        response=response.answer,
    ),),
)
```

2.4 Peer context and durable gates

For rounds 1 and 2, `previous.submissions.for_current_participant(...)` creates a serialized, participant-relative view. The prompt injects its `self_response` and `peer_responses` fields separately. Participant identities remain available for auditing, but the recipient no longer has to recover its own response from an undifferentiated batch.

The report directory's parent also contains a frozen `.ep-workflow` audit log. Its ordered gates require the design artifact to exist and the experiment tests to pass. When pilot findings changed the implementation, the earlier gates were cleared and reverified rather than silently retaining stale evidence.

3 Local OpenAI setup and execution

The pilot uses EDSL locally for orchestration and SQLite persistence while making direct OpenAI API calls. "Local" here does not mean an on-device language model.

```
answerer = EDSLAgentAnswerer(
    Model("gpt-4o-mini"),
    run_options={
        "disable_remote_inference": True,
        "disable_remote_cache": True,
        "cache": False,
        "stop_on_exceptions": True,
    },
)
WorkflowSimulation(
    coordinator,
    {agent.name: agent for agent in agents},
    answerer,
).run(instance_id)
```

Reproduce the pilot from the repository root:

```
ep auth status
python -m examples.talk_isnt_cheap.run_openai_pilot \
  --output-dir examples/talk_isnt_cheap/runs/my-openai-pilot \
  --model gpt-4o-mini
```

The run produces `workflow.sqlite`, `shared-state.sqlite`, and `summary.json`. The two databases retain orchestration events and atomic state history; the JSON file is the portable analysis artifact.

4 Pilot results

	Independent (0)	Debate 1	Debate 2
Agent 1	B	B	B
Agent 2	B	B	B
Agent 3	B	B	B
Majority	B	B	B
Reference answer	D	D	D
Majority correct	No	No	No

Table 1: All nine validated responses in the final GPT-4o-mini pilot.

Across the two adjacent round pairs and three participants there are six transitions:

Transition	Count
Correct → correct	0
Incorrect → correct	0
Correct → incorrect	0
Incorrect → incorrect	6

Every transition occurred while both peers agreed with the participant's previous answer. Thus the pilot illustrates unanimous persistence, not persuasion-induced flipping. The rationales were also highly similar: agents interpreted "doesn't last" as gradual wear and contrasted that with the more abrupt "falling apart." Debate reinforced this shared interpretation without introducing new evidence.

Compact transcript

Stage	Ans.	Representative rationale
Round 0	B	A product that does not last typically wears out over time; the phrase suggests gradual degradation rather than immediate failure.
Round 1	B	Peer responses agree that wearing out captures declining durability; disintegration and falling apart appear more abrupt.
Round 2	B	Wearing out continues to be preferred because it describes gradual loss of quality and functionality through regular use.

5 Comparison with the paper

Dimension	Original paper	EDSL pilot
Scope	3 datasets; 100 items each; 5 seeds; 10 model compositions	1 CSQA item; 1 run; 1 composition
Models	GPT-4o-mini, LLaMA-3.1-8B, Mistral-7B	3 × GPT-4o-mini
Rounds	Independent response plus 2 debate rounds	Same
Response	Answer plus reasoning in one generation	Typed answer plus reasoning in one generation
Peer evidence	Own prior response plus other agents' prior responses	Identity-preserving preceding-round batch
Persistence	Paper reports aggregate patterns, including harmful flips	All agents retained B through both revisions
Comparable CSQA result	3-GPT group: 75.6% before, 74.8% after	0/1 before, 0/1 after
Inference	Full empirical comparison	Pipeline smoke test only
Auditability	Research code and aggregate figures	Serializable workflow, event store, state history, JSON summary, frozen gates

The single item is directionally compatible with the paper's warning that discussion need not fix an initially wrong group, but it does not demonstrate the paper's signature harmful transition from correct to incorrect. Nor can a one-item result be compared numerically with a 100-item, five-seed accuracy estimate.

6 What a full replication still requires

- Acquire and version the exact CSQA, MMLU, and GSM8K samples for all five seeds.
- Bind each participant seat to the intended model for every homogeneous and heterogeneous composition; verify contemporary provider identifiers for the two open-weight models.
- Run both base and correctness-payoff prompt conditions with the paper's decoding settings.
- Define and document tie/filtering behavior, numeric normalization for GSM8K, failed-call retries, and any context summarization threshold.

- Aggregate accuracy by round, transition classes, undesirable flips by peer agreement, model, dataset, treatment, composition, and seed; report uncertainty across seeds.
- Treat this pilot's unanimous B response as diagnostic data, not evidence that the reference key is wrong; verify every imported benchmark answer against the pinned dataset artifact.

Artifacts and provenance

Design	`examples/talk_isnt_cheap/debate_experiment.py`
Runner	`examples/talk_isnt_cheap/run_openai_pilot.py`
Final pilot	`examples/talk_isnt_cheap/runs/openai-gpt-4o-mini-pilot-4/`
Tests	`tests/examples/test_talk_isnt_cheap.py`
Source paper	Wynn, Andrea; Satija, Harsh; Hadfield, Gillian K. (2025), arXiv:2509.05396v2

Generated from the saved EDSL design and final pilot artifacts. No model was used to score or rewrite the saved answers.